

Aaron Weymouth

① Detailed presentation of coulomb

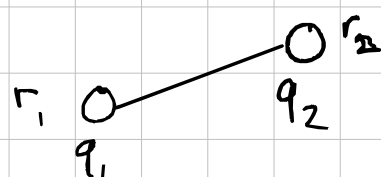
② 1.2

③ 1.3

for coulomb: let there be 2 point charges: q_1, q_2 at r_1, r_2

let $\vec{r}_{12} = r_1 - r_2$ and $\hat{r}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|}$

$$\hat{r}_{12} = \frac{\vec{r}_{12}}{r_{12}}$$



Coulombs law says the force on q_1 due to q_2 is

$$F_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \left(\frac{r_1 - r_2}{|r_1 - r_2|^3} \right)$$

Thus $q_1 q_2 > 0 \Rightarrow$ repulsive, $q_1 q_2 < 0 \Rightarrow$ attractive

Also: $F_{21} = -F_{12}$; so force is central and satisfies Newtons 3rd

Lets define: $\vec{E}(x)$ electric **field** as force per unit test charge

$E(\vec{r}) = \vec{F}/q_0$; hence a point charge q at r' produces

$$E(\vec{r}) = \frac{1}{4\pi\epsilon_0} \cdot q \frac{(r - r')}{|r - r'|^3} \quad \text{and the test charge } q_0 \text{ Experiences}$$
$$\vec{F} = q_0 \vec{E}$$

• for Extending to many point charges - fields add by Superposition

$$\vec{E}(r) = \frac{1}{4\pi\epsilon_0} \sum_i q_i \frac{r - r_i}{|r - r_i|^3} \quad \text{and for continuous charge distribution}$$

$$dq = \rho(r') d^3 r' \rightarrow \vec{E}(r) = \frac{1}{4\pi\epsilon_0} \int \rho(r') \frac{r - r'}{|r - r'|^3} d^3 r'$$

Which is the continuous form of coulombs law.

The Electrostatic field is conservative thus:

$$\nabla \times \vec{E} = 0 \quad ; \quad \text{so we may define a scalar potential}$$

$$\text{Such that } \vec{E} = -\nabla \phi$$

and back to the continuous charge density:

$$\phi(r) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(r')}{|r - r'|} d^3 r' \quad \text{and therefore the coulomb law may be expressed as either } \vec{E} \text{ or } \phi$$

finally, using gauss law:

$$\nabla \cdot \vec{E} = \rho/\epsilon_0 \quad \text{and } \vec{E} = -\nabla \phi \quad \text{then our laplacian}$$

$$-\nabla^2 \phi = \rho/\epsilon_0 \quad \text{or } \nabla^2 \phi = -\frac{\rho}{\epsilon_0} \quad \left[\text{which is 'poisson's'} \right]$$

Overall, coulombs law gives the electrostatic interaction between point charges and through Superposition, determines the electric **field** and potential for arbitrary Static charge distribution!

Problem 1.2:

given orthogonal coordinates

$$ds^2 = \frac{du^2}{u^2} + \frac{dv^2}{v^2} + \frac{dw^2}{w^2}, \quad d^3x = \frac{du dv dw}{uvw}$$

$$\text{Recall: } \int^3 (x-x') = \lim_{\alpha \rightarrow 0} \frac{e^{-|x-x'|^2 / 2\alpha^2}}{(2\pi)^{3/2} \alpha^3}$$

so near x' :

$$|x-x'|^2 \simeq \frac{(u-u')^2}{u'^2} + \frac{(v-v')^2}{v'^2} + \frac{(w-w')^2}{w'^2}$$

$$\begin{aligned} \text{Thus: } \int^3 (x-x') &= \lim_{\alpha \rightarrow 0} \prod_{\substack{u,v,w \\ =q}} \frac{1}{\sqrt{2\pi} \alpha} e^{-(q-q')^2 / 2\alpha^2 q'^2} \\ &= u'v'w' \prod_{q=u,v,w} \lim_{\alpha \rightarrow 0} \frac{1}{\sqrt{2\pi} \alpha q'} e^{-(q-q')^2 / 2\alpha^2 q'^2} \end{aligned}$$

$$\text{Recall: } \lim_{\epsilon \rightarrow 0} \frac{1}{\sqrt{2\pi} \epsilon} e^{-x^2 / 2\epsilon^2} = \delta(x)$$

$$\hookrightarrow \int^3 (x-x') = u'v'w' \delta(u-u') \delta(v-v') \delta(w-w')$$

let's set $x=x'$ thus $u'v'w' = u.v.w$

$$\rightarrow \int^3 (x-x^3) = UVW \delta(u-u') \delta(v-v') \delta(w-w')$$

$$\hookrightarrow \text{check } \int UVW \delta(u-u') \delta(v-v') \delta(w-w') \cdot \frac{du dv dw}{uvw} = 1$$

P. 1.3:

We use $Q = \int \rho d^3x$; let s be cylindrical radius

(a) spherical shell radius R

$$\text{let } \rho = A \delta(r-R)$$

$$\begin{aligned} \text{then } Q &= A \int_0^\infty \delta(r-R) r^2 dr \cdot \int_0^\pi \sin\theta d\theta \cdot \int_0^{2\pi} d\phi \\ &= A(R^2)(2)(2\pi) = 4\pi(R^2)(A) = 4\pi R^2 A = Q \end{aligned}$$

$$\hookrightarrow A = \frac{Q}{4\pi R^2}$$

$$\therefore \rho(r, \theta, \phi) = \frac{Q}{4\pi R^2} \delta(r-R)$$

(b) cylindrical shell radius b . λ is charge per unit length.

$$\text{let } \rho = A \delta(s-b)$$

$$\text{then } \lambda = A \cdot \int_0^{2\pi} d\phi \int_0^\infty s \delta(s-b) ds = 2(2\pi) \cdot b$$

$$\hookrightarrow A = \frac{\lambda}{2\pi b}$$

$$\therefore \rho(s, \phi, z) = \frac{\lambda}{2\pi b} \delta(s-b)$$

c. (c) flat disk; cylindrical: the disk is at $z=0$, $0 \leq s \leq R$

$$\text{let } \rho = A \delta(z) \Theta(R-s)$$

$$\begin{aligned} \text{then } Q &= A \cdot \int_{-\infty}^\infty \delta(z) dz \cdot \int_0^{2\pi} d\phi \cdot \int_0^R s ds \\ &= A \cdot [(1)(\pi)(R^2)] = A\pi R^2 = Q \end{aligned}$$

$$\hookrightarrow A = \frac{Q}{\pi R^2}$$

$$\Rightarrow \rho(s, \phi, z) = \frac{Q}{\pi R^2} \delta(z) \Theta(R-s)$$

(d) flat disc; spherical coordinates: recall $z = r \cos\theta$

$$\text{in part c: } \rho = \frac{Q}{\pi R^2} \delta(r \cos\theta) \Theta(R - r \sin\theta) \quad s = r \sin\theta$$

$$\text{Also: } \delta(f(\theta)) = \sum_i \frac{\delta(\theta - \theta_i)}{|f'(\theta_i)|}$$

$$\text{here: } f(\theta) = r \cos\theta; \theta_i = \pi/2; |f'(\pi/2)| = r$$

$$\text{thus: } \delta(r \cos\theta) = \frac{1}{r} \delta(\theta - \frac{\pi}{2})$$

$$\text{on } \theta = \pi/2; \underline{r \sin\theta = r}$$

$$\therefore \rho(r, \theta, \phi) = \frac{Q}{\pi R^2 r} \delta(\theta - \pi/2) \Theta(R-r)$$

$$\text{check: } \int \rho d^3x = \frac{Q}{\pi R^2} \int_0^{2\pi} d\phi \int_0^R r dr \int_0^\pi \delta(\theta - \frac{\pi}{2}) \sin\theta d\theta$$

$$= \frac{Q}{\pi R^2} (2\pi) \left(\frac{R^2}{2}\right) (1)$$

$$= Q \quad \checkmark$$